B K Birla Institute of Engineering & Technology, Pilani

45 Days Summer Training Program
Subject: Data Science with AIML

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Assignment 2

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Question 1.
Create a tuple.
Print:
First element
Last element
Length of the tuple
Elements from index 1 to 3

Sol.

```
my_tuple = (10, 20, 30, 40, 50)

print("First Element:", my_tuple[0])
print("Last Element:", my_tuple[-1])
print("Length of Tuple:", len(my_tuple))
print("Elements from Index 1 to 3:", my_tuple[1:4])
```

Output:
First Element: 10
Last Element: 50
Length of Tuple: 5
Elements from Index 1 to 3: (20, 30, 40)

Question 2:
Create a tuple.
Print:
Second fruit
Last two fruits

Total number of fruits

Solution:

```
fruits = ("apple", "banana", "mango", "orange")

print("Second Fruit:", fruits[1])
print("Last Two Fruits:", fruits[-2:])
print("Total Number of Fruits:", len(fruits))
```

Output:
Second Fruit: banana
Last Two Fruits: ('mango', 'orange')
Total Number of Fruits: 4

Question 3.
Create a set.
Print:
Complete set
Length of the set
Check whether 30 is present in the set

Solution:

```
numbers = {10, 20, 30, 40, 50}

print("Complete Set:", numbers)
print("Length of Set:", len(numbers))

if 30 in numbers:
    print("30 is present in the set")
else:
    print("30 is not present in the set")
```

Output:

Complete Set: {10, 20, 30, 40, 50}
Length of Set: 5
30 is present in the set

Question 4:
Create two sets.

Find:
Union
Intersection

Solution:

```
set1 = {1, 2, 3, 4}
set2 = {3, 4, 5, 6}

union_set = set1.union(set2)
intersection_set = set1.intersection(set2)

print("Union:", union_set)
print("Intersection:", intersection_set)
```

Output:

```
Union: {1, 2, 3, 4, 5, 6}
Intersection: {3, 4}
```

Question 5:
Create a dictionary.
Print:
Complete dictionary
Student name
Student age
Student course

Solution:

```
student = {
    "name": "Praveen",
    "age": 20,
    "course": "Python"
}

print("Complete Dictionary:", student)
print("Student Name:", student["name"])
print("Student Age:", student["age"])
print("Student Course:", student["course"])
```

Output:

Complete Dictionary: {'name': 'Praveen', 'age': 20, 'course': 'Python'}
Student Name: Praveen
Student Age: 20
Student Course: Python

Question 6:
Create a list
Write a program to find:
Largest element
Second largest element
Smallest element

Solution:

```
numbers = [12, 45, 7, 23, 56, 89, 34]

largest = numbers[0]
smallest = numbers[0]

for num in numbers:
    if num > largest:
        largest = num

    if num < smallest:
        smallest = num

second_largest = smallest

for num in numbers:
    if num > second_largest and num != largest:
        second_largest = num

print("Largest Element:", largest)
print("Second Largest Element:", second_largest)
print("Smallest Element:", smallest)
```

Output:

```
Largest Element: 89
Second Largest Element: 56
Smallest Element: 7
```

Question 7:
Create a list.
Write a function that takes the list as input and returns the list in reverse order without using the reverse() method.

Solution:

```
def reverse_list(arr):
    reversed_arr = []

    for i in range(len(arr)-1, -1, -1):
        reversed_arr.append(arr[i])

    return reversed_arr

arr = [10, 20, 30, 40, 50, 60]

print("Original List:", arr)
print("Reversed List:", reverse_list(arr))
```

Output:

```
Original List: [10, 20, 30, 40, 50, 60]
Reversed List: [60, 50, 40, 30, 20, 10]
```

Question 8:
Create a tuple.
Find:
Count of elements divisible by 5
Sum of all elements
Average of the tuple

Solution:

```
data = (5, 10, 15, 20, 25, 30, 35)

count = 0
total = 0

for num in data:
    if num % 5 == 0:
        count += 1
```

```
    total += num

average = total / len(data)

print("Count of Elements Divisible by 5:", count)
print("Sum of Elements:", total)
print("Average:", average)
```

Output:

```
Count of Elements Divisible by 5: 7
Sum of Elements: 140
Average: 20.0
```

Question 9:
Create a dictionary.
Find:
Student with highest marks
Student with lowest marks
Students scoring more than 85 marks

Solution:

```
students = {
    "Aman": 78,
    "Riya": 92,
    "Kriti": 88,
    "Rahul": 95
}

highest_name = ""
lowest_name = ""

highest_marks = -1
lowest_marks = 101

for name, marks in students.items():

    if marks > highest_marks:
        highest_marks = marks
        highest_name = name

    if marks < lowest_marks:
```

```
        lowest_marks = marks
        lowest_name = name

print("Highest Scorer:", highest_name, "-", highest_marks)
print("Lowest Scorer:", lowest_name, "-", lowest_marks)

print("\nStudents Scoring More Than 85:")

for name, marks in students.items():
    if marks > 85:
        print(name, "-", marks)
```

Output:

```
Highest Scorer: Rahul - 95
Lowest Scorer: Aman - 78

Students Scoring More Than 85:
Riya - 92
Kriti - 88
Rahul - 95
```

Question 10:
Write a function.
that takes a list as input and prints the frequency of each element.

Solution:

```
def count_frequency(arr):

    frequency = {}

    for item in arr:
        if item in frequency:
            frequency[item] += 1
        else:
            frequency[item] = 1

    for key, value in frequency.items():

        if value == 1:
            print(key, "->", value, "time")
        else:
```

```python
            print(key, "->", value, "times")

arr = [1, 2, 2, 3, 1, 4, 2]

count_frequency(arr)
```

Output:

1 -> 2 times
2 -> 3 times
3 -> 1 time
4 -> 1 time

Question 11:
Write a function.
that takes a list as input and prints all duplicate elements present in the list.

Solution:

```python
def find_duplicates(arr):

    seen = set()
    duplicates = set()

    for item in arr:

        if item in seen:
            duplicates.add(item)
        else:
            seen.add(item)

    print("Duplicate Elements are:")

    for item in duplicates:
        print(item)

arr = [10, 20, 30, 20, 40, 10, 50, 30]

find_duplicates(arr)
```

Output:

Duplicate Elements are:

10
20
30